# Lecture 22: Summary of Curve Sketching

**Goal:** Use the tools of calculus to draw a more accurate graph of a function.
All examples come from §4.5 of James Stewart's *Calculus Early Transcendentals*, 5th Edition as well as §4.4 of *Calculus Early Transcendentals*, 3rd Edition by Briggs et. al. (I'm too lazy to make my own examples).

All graphs will be available on Desmos (here).

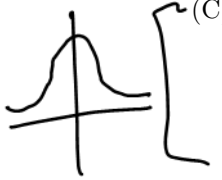
## Guidelines for Sketching a Curve

For drawing the graph of $y = f(x)$ by hand, consider:

(A) **Domain:** Find all values where $f$ is defined.

(B) **Intercepts:** Find the $y$-intercept by computing $f(0)$. Find the $x$-intercepts by solving the equation $f(x) = 0$.

(C) **Symmetry:**

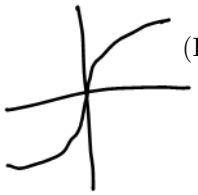


(i) *Even functions:* When $f(x) = f(-x)$; this means the graph is symmetric about the $y$-axis.

(ii) *Odd functions:* When $f(-x) = -f(x)$; this means it is symmetric about the origin (the left side is a 180° rotation of the right side)

(iii) *Periodic functions:* When $f(x+p) = f(x)$ for some $p > 0$, the smallest such $p$ is called the period. You only need to focus on one period of the function.

(D) **Asymptotes and End Behavior:**



(i) *Vertical Asymptotes:* Occurs at $x = a$ when either:

$$\lim_{x \to a^-} f(x) \text{ OR } \lim_{x \to a^+} f(x)$$

is not finite (equals $\infty$ or $-\infty$).

(ii) *Horizontal Asymptotes:* Occurs at $y = L$ when either:

$$\lim_{x \to -\infty} f(x) = L \quad \text{OR} \quad \lim_{x \to \infty} f(x) = L$$

(iii) *Slant Asymptotes:* The line $y = mx + b$ is a slant asymptote when either :

$$\lim_{x \to -\infty} [f(x) - (mx + b)] = 0 \quad \text{OR} \quad \lim_{x \to \infty} [f(x) - (mx + b)] = 0$$

For rational functions, if the quotient from long division $q(x)$ is linear, then $q(x)$ is the slant asymptote.

(E) **Intervals of Increase or Decrease:** Use the I/D Test to see when $f$ is increasing or decreasing. If $f' > 0$, then $f$ is increasing. If $f' < 0$, $f$ is decreasing.

(F) **Relative Maximum and Minimum Values:** Use the 1st or 2nd derivative test to determine if critical numbers are relative extrema.

(G) **Concavity and Inflection Points:** Use the concavity test. If $f'' > 0$, then $f$ is concave upwards. If $f'' < 0$, $f$ is concave downwards.

(H) **If needed, get more information:** You can also compute $(x, f(x))$ pairs to see the height of the graph. The tangent line also tells you where the graph is going at that point.

**Example 1** (Why Graphing Calculators Can Be Deceiving)**.** Sketch a graph of $f(x) = \frac{1}{3}x^3 - 400x$

Domain: $(-\infty, \infty) = \mathbb{R}$

End Behavior:

$$\lim_{x\to\infty}\left(\frac{1}{3}x^3 - 400x\right) = \lim_{x\to\infty}\frac{1}{3}x^3 = \infty$$

$$\lim_{x\to-\infty}\left(\frac{1}{3}x^3 - 400x\right) = \lim_{x\to-\infty}\frac{1}{3}x^3 = -\infty$$

Intercepts:

y-int: $f(0) = \frac{1}{3}\cdot 0^3 - 400\cdot 0 = 0$

x-int: $\frac{1}{3}x^3 - 400x \overset{\text{set}}{=} 0$

$x^3 - 1200x = 0$

$x(x^2 - 1200) = 0$

$x = 0, \pm\sqrt{1200}$

$\approx 0, \pm 34.6$



$\Longrightarrow$

ID and C Test

$f(x) = \frac{1}{3}x^3 - 400x$

$f'(x) = x^2 - 400 \overset{\text{set}}{=} 0$

$x^2 = 400$

$x = \pm 20$

$f''(x) = 2x \overset{\text{set}}{=} 0 \Rightarrow x = 0$

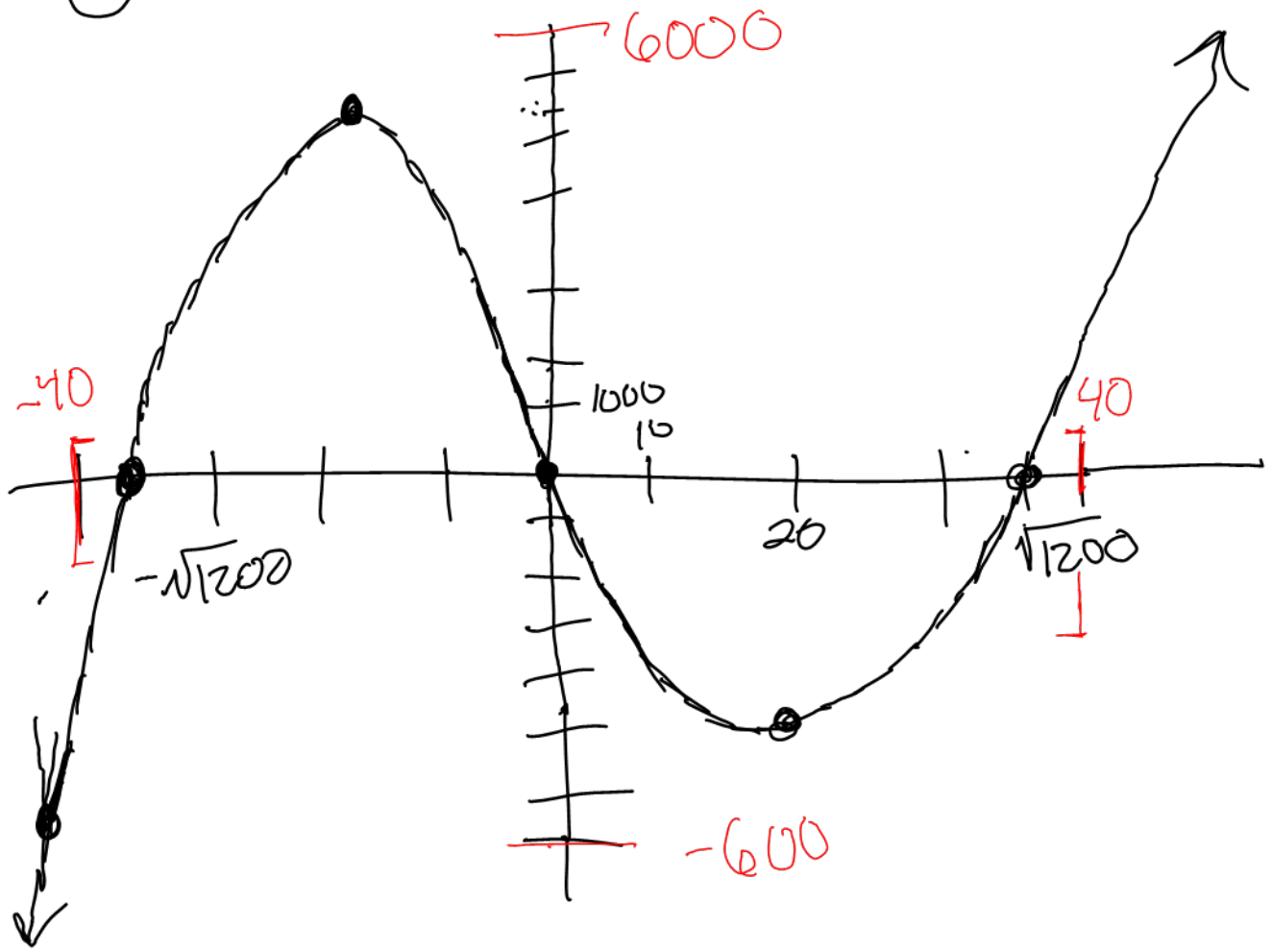
| Test Point | $-100$ | $-20$ | $-1$ | $0$ | $1$ | $20$ | $100$ |
|---|---|---|---|---|---|---|---|
| Sign of $f'$ | $+$ | | $-$ | | $-$ | | $+$ |
| Sign of $f''$ | $-$ | | $-$ | | $+$ | | $+$ |
| 1D Test | Inc ↗ | | Dec ↘ | | Dec ↘ | | Inc ↗ |
| C Test | CD | | CD | | CU | | CU |

Rel. Ext. & IPs:

① Rel. max at $(-20, f(-20)) = (-20, 5333\frac{1}{3})$

② Rel. mins at $(20, f(20)) = (20, -5333\frac{1}{3})$

③ IP at $(0, f(0)) = (0, 0)$



**Example 2.** Sketch a graph of $f(x) = \dfrac{2x^2}{x^2-1}$

VAs: $x^2-1 \overset{\text{set}}{=} 0$

$\Rightarrow x = \pm 1$

VAs at $x = \pm 1$

Domain: $(-\infty,-1) \cup (-1,1) \cup (1,\infty) = \mathbb{R} - \{-1,1\}$

HAs:

$$\lim_{x\to\infty} \frac{2x^2}{x^2-1} = \lim_{x\to\infty} \frac{2x^2}{x^2} = 2$$

We have a HA at $y = 2$

Intercepts: $y$-int: $f(0) = \dfrac{2\cdot 0^2}{0^2-1} = 0$

$x$-int: $2x^2 = 0 \Rightarrow x = 0$

I.e. $f$ only crosses the origin

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I/D and C Test:

$$f(x) = \frac{2x^2}{x^2-1}$$

$$f'(x) = \frac{4x(x^2-1) - 2x^2(2x)}{(x^2-1)^2}$$

$$= \boxed{\frac{-4x}{(x^2-1)^2}} \overset{\text{set}}{=} 0$$

$$-4x = 0$$

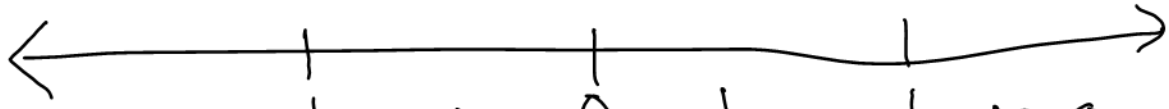
$$x = 0$$

$$f''(x) = \frac{-4(x^2-1)^2 - (-4x)(2)(x^2-1)(2x)}{(x^2-1)^4}$$

$$= \frac{-4(x^2-1) + 16x^2}{(x^2-1)^3} = \frac{-4x^2+4+16x^2}{(x^2-1)^3}$$

$$= \frac{12x^2+4}{(x^2-1)^3}$$

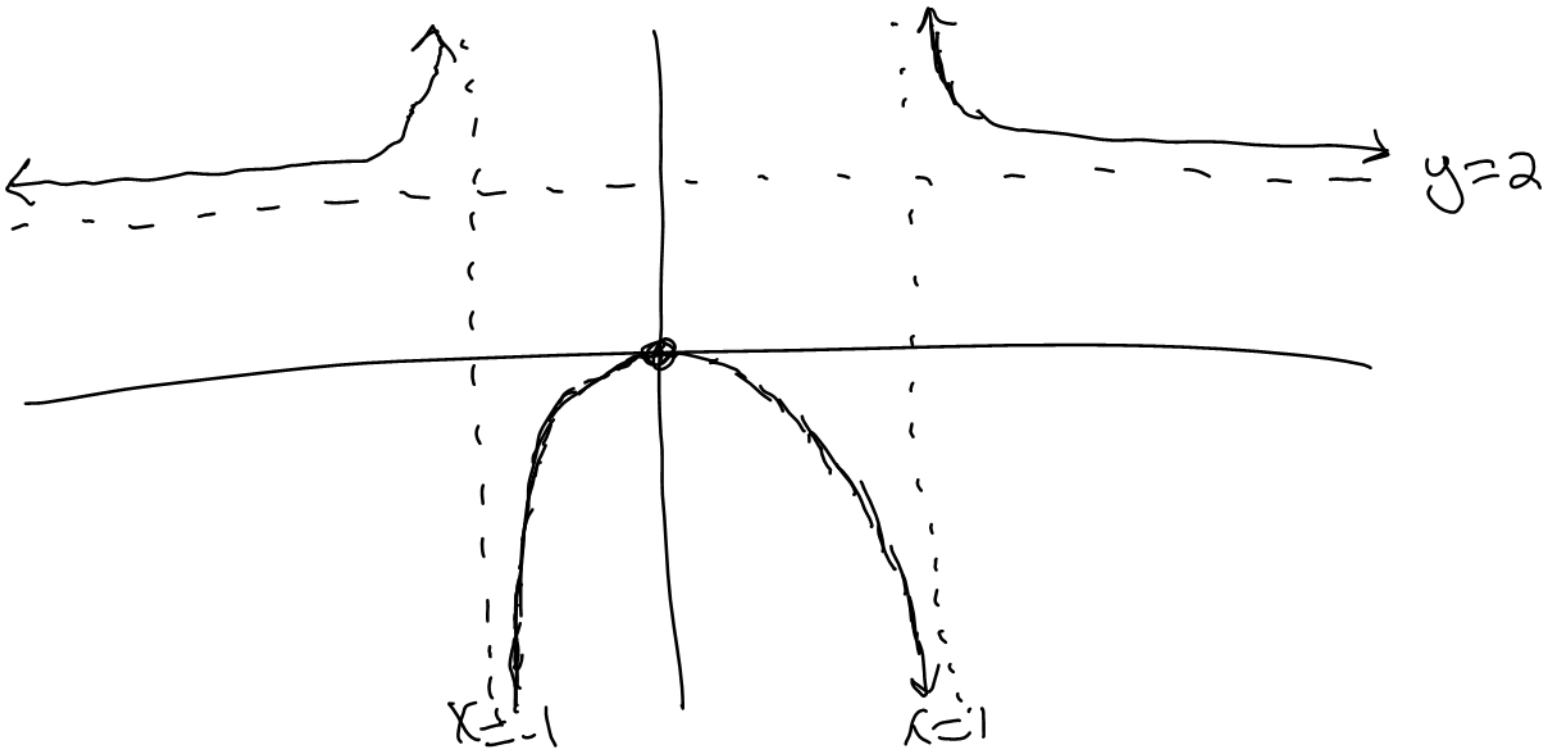
Since concavity could change at the asymptotes, include them in the sign chart

$-4x$ 

| Test Point | $-100$ | $-1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $1$ | $100$ |
|---|---|---|---|---|---|---|---|
| Sign of $f'(x)$ ☆ | + | | + | | − | | − |
| Sign of $12x^2+4$ | + | | + | | + | | + |
| Sign of $(x^2-1)^3$ | + | | − | | − | | + |
| Sign of $f''$ | + | | − | | − | | + |
| ID Test | Inc | | Inc ↗↘ | | Dec | | Dec |
| C Test | CU | | CD | | CD | | CU |

① Relative max at $(0, f(0)) = (0,0)$

② C changes at $x = -1$ and $x = 1$


y=2
x=-1
x=1

**Example 3.** Sketch a graph of $f(x) = \dfrac{x^3}{x^2+1}$

$\dfrac{x^3}{x^2+1}$

Intercepts: Only passes through the origin

VAs: $x^2 + 1 = 0$

NO SOLUTION

HAs: No HAs b/c there is a slant asymptote

$$x^2+1 \overline{)\, x^3}$$

$x$ ← $y = x$ is our slant asymptote

$$-(x^3 + x)$$

$$-x$$

$$f(x) = \frac{x^3}{x^2+1}$$

$$f'(x) = \frac{3x^2(x^2+1) - x^3(2x)}{(x^2+1)^2}$$

$$= \frac{x^4 + 3x^2}{(x^2+1)^2}$$

$$= \frac{x^2(x^2+3)}{(x^2+1)^2} \quad \begin{matrix} \geq 0 \\ \geq 0 \\ \geq 0 \end{matrix}$$

① Only 0 at $x=0$

② $f$ is always increasing by $1^{st}$ $\frac{d}{dx}$ test

**Example 4.** Sketch a graph of $f(x) = \dfrac{10x^3}{x^2 - 1}$

**Example 5** (Bell Curve)**.** Sketch a graph of $f(x) = e^{-x^2}$

**Example 6.** Sketch a graph of $f(x) = \ln(4 - x^2)$